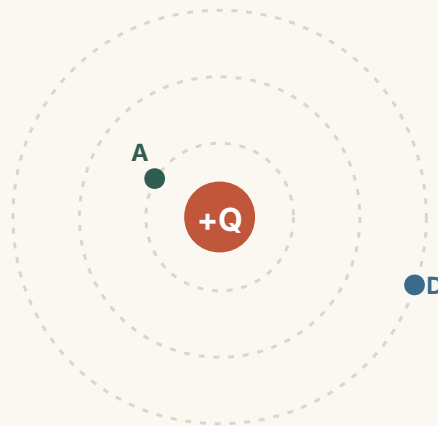


Electric Potential

From the energy of one charge to a property of the field itself.

ESSENTIAL QUESTION

What is electric potential — and how is it different from potential energy and field strength?



● LESSON MAP

Build potential in four moves.

01

STORE

The electric force is conservative, so a charge has potential energy.

02

DEPEND

That energy grows with the charge — it isn't the field alone.

03

DIVIDE

Divide energy by charge to get potential, a field property.

04

READ

Use potential to compare points and tell it from E.

KEEP THIS IN MIND

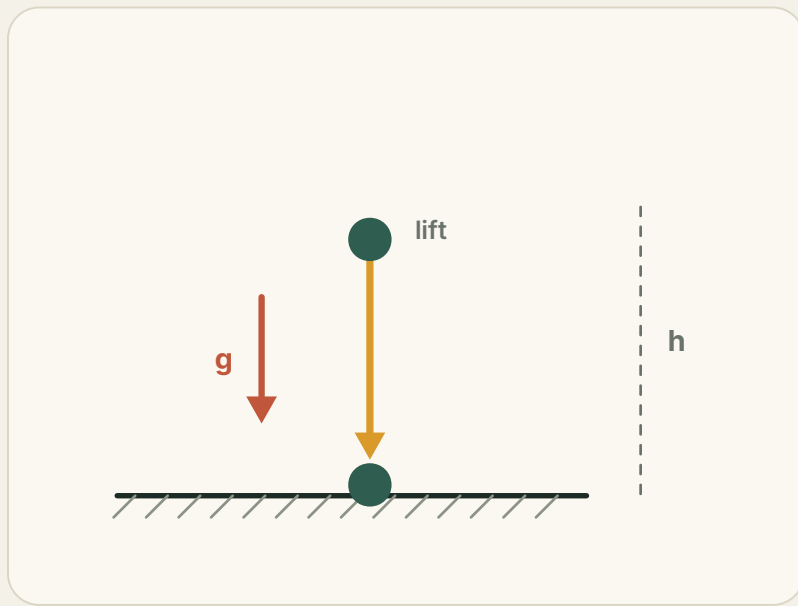
Potential **energy** belongs to a charge. **Potential** belongs to the field.

RECALL

Gravity already taught us this idea.

Lift a ball and gravity stores energy you can get back — no matter which path you take.

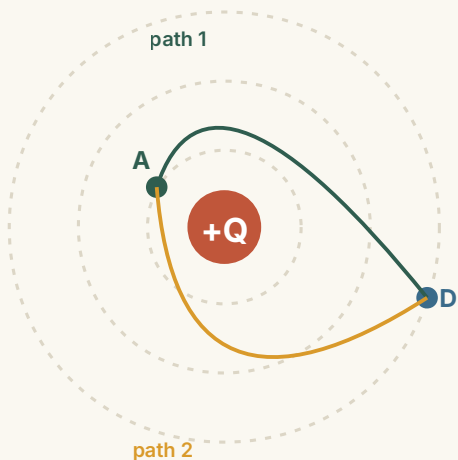
- 1 Because gravity is conservative**
PE depends only on height, not on the path taken.
- 2 Work sets the change**
The work done fixes the energy change, W .



OBSERVE

Moving a charge: the path doesn't matter.

Around a source charge $+Q$, the work to carry a charge from A to D is the same for every path.



WHAT THIS MEANS

The electric force is **conservative**: work depends only on the start and end points.

So we can give each point an electric potential energy — the energy a charge stores sitting there.

MODEL

Work by the field = drop in potential energy.

Exactly like gravity: when the field does positive work, stored energy falls.

$$W_{AB} = E_{pA} - E_{pB}$$

POSITIVE WORK

The field pushes the charge along —
potential energy **decreases**.

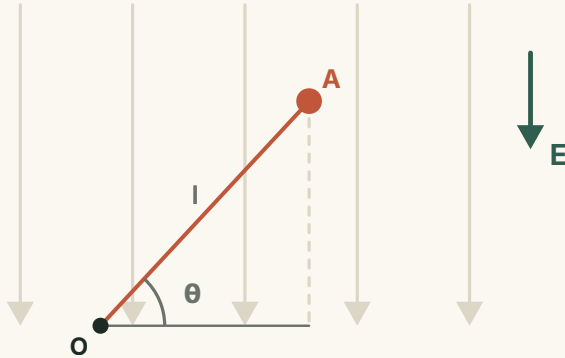
NEGATIVE WORK

You push against the field — potential
energy increases.

THE CATCH

But this energy depends on the charge.

In a uniform field E , a charge at point A (distance l , angle θ) stores energy that scales with q .



ENERGY STORED

$$E_p = qEl \cos \theta$$

Double the charge $\rightarrow$ double the energy. So E_p describes the **charge**, not just the field.

KEY INSIGHT

Divide out the charge — what's left is the field's.

For every test charge at the same point, the ratio E_p / q comes out the same.

$$\frac{E_p}{q} = \text{same for any charge} \Rightarrow \varphi = \frac{E_p}{q}$$

This ratio is a property of the point in the field — the electric potential φ .

Two field properties, built the same way.

Each one divides a charge's experience by the charge itself.

FIELD STRENGTH · FORCE PER CHARGE

$$E = \frac{F}{q}$$

Vector. How hard the field pushes one unit of charge.
Units: N C^{-1}

POTENTIAL · ENERGY PER CHARGE

$$\varphi = \frac{E_p}{q}$$

Scalar. How much PE one unit of charge has. Units: V
 $= \text{J C}^{-1}$

PROPERTIES

What potential tells you about a point.

It is a scalar attached to the field, fixed once you choose a zero.

Sign

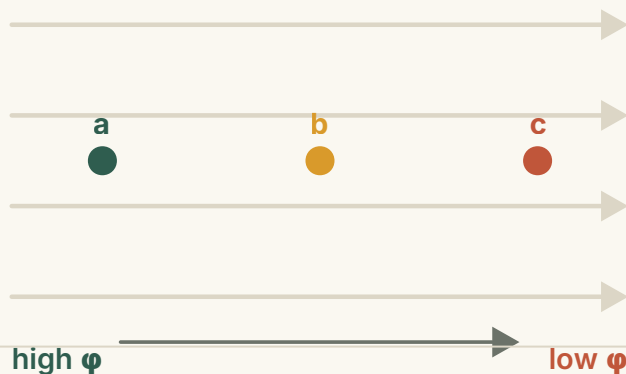
Take zero at infinity: potential is positive near +, negative near -.

Falls along field lines

Move along a field line and potential drops — the field points downhill.

Reference

Only differences matter; the zero point is a choice — infinity or earth.



READ IT

Field lines point from high to low potential.

Denser lines mean a stronger field E ; following a line, the potential always decreases.



A field line is a one-way street for potential: step along it and ϕ can only **fall**.

THE RULE

Stronger field $\rightarrow$ steeper potential drop over the same distance.

DON'T CONFUSE THEM

Three quantities, three questions.

Two belong to the field; one belongs to a particular charge.

Field strength

Field · vector. Force per unit charge
— how strong the push is.

Potential

Field · scalar. Energy per unit charge
at a point.

Potential energy

The charge's own. Energy this
particular charge stores.

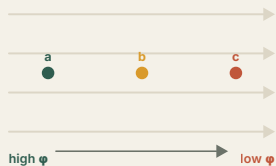
TELL THEM APART Ask: per charge, or for this charge? Field property, or particle property?

● SUMMARY

Potential energy belongs to a charge; divide by the charge for **potential**.

POTENTIAL — ENERGY PER UNIT CHARGE

$$\varphi = \frac{E_p}{q}$$



A scalar map of the field itself, just as E maps the force.

Follow a field line and φ always falls.